

Vehicle Sideslip Angle Estimation Using a Zonotopic Kalman Filter Under Modeling and Measurement Uncertainties

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Abstract—The vehicle sideslip angle is a critical factor in ensuring safe and stable vehicle operation; however, it cannot be measured directly and easily during normal driving. Conventional estimation methods frequently disregard model and measurement uncertainties, offering point estimates that lack a quantitative assessment of their reliability. In order to address these limitations, the present paper proposes a sideslip angle estimator based on a zonotopic Kalman filter. This approach provides both an estimate and a guaranteed bound that accounts for system uncertainties. The approach under consideration explicitly models and propagates measurement noise and parametric uncertainties, resulting in bounded, reliable estimates over time. The validation process is executed through the implementation of co-simulations, employing CarSim, a high-fidelity vehicle dynamics simulator, which is integrated with MATLAB. The findings indicate that the estimated zonotopes consistently envelop the ground truth sideslip angle across a range of driving scenarios, thereby demonstrating the robustness and reliability of the estimator.

Index Terms—vehicle sideslip angle estimation, zonotope, kalman filter, uncertainty, vehicle safety.

I. INTRODUCTION

During the last decades, a wide range of driving automation systems have been incorporated into series-production vehicles, yielding substantial advantages in terms of driving safety and comfort [1]–[3].

Among the various control techniques employed, many rely critically on accurate knowledge of the vehicle's sideslip angle. This state is necessary for functions such as advanced path and

trajectory tracking [4]–[7]. Nevertheless, direct measurement of the sideslip angle using onboard sensors is not currently feasible [8]. Specifically, while the measurement of the sideslip angle can be conducted through optical sensors [9], these are very expensive and therefore not viable for implementation in series-production vehicles. Consequently, the sideslip angle must typically be estimated in real-time using a combination of vehicle dynamics models, sensor fusion, and observer-based or data-driven algorithms [10]. The development of robust estimation methods for varying driving conditions remains a critical area of research in vehicle control and automation [11].

Up to date, very mature and well-known techniques have been used for sideslip angle estimation [12]–[16]. In [17], an extended Kalman filter that uses a rational tyre model is presented to estimate the sideslip angle in heavy vehicles. In [18], low-cost GPS and IMU sensors are used to estimate the sideslip angle. In [19], an automated tuned Unscented Kalman Filter for vehicle sideslip angle estimation is proposed. In [20], dynamic neural networks are used to predict the sideslip angle. Despite the abundance of relevant literature addressing the estimation of sideslip angle, it is uncommon to provide the associated estimation uncertainty, caused by various factors, including sensor noise or model uncertainties [21], [22]. As an example, [23] presents an interval observer design for LPV systems with parametric uncertainty, with a validation on slip angle estimation.

With respect to techniques for uncertainty analysis and propagation, zonotopes (and zonotopic-based estimation [24]) are gaining increasing relevance due to their favorable mathematical properties and computational efficiency [25]. Zonotopes, a distinct class of convex polytopes, are particularly

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well-suited for representing bounded uncertainty in high-dimensional spaces [26]. Their algebraic structure enables the efficient execution of set operations, including Minkowski addition and linear transformations [27]. These operations are critical in various fields, such as reachability analysis, robust control, and estimation tasks [28]–[31]. In contrast to other set representations, such as ellipsoids or general polytopes, zonotopes exhibit a unique balance between expressiveness and tractability. They possess the capacity to tightly enclose complex uncertainty regions while facilitating efficient computation. This renders them particularly well-suited for real-time applications, including autonomous vehicle motion planning, safety verification, and model predictive control under uncertainty [32]. Moreover, recent advancements have expanded the utilization of these methods in nonlinear systems through linearization techniques or hybrid representations, thereby broadening their applicability across domains where uncertainty quantification is paramount [33].

For all the aforementioned reasons, this paper proposes a zonotopic Kalman filter that estimates the sideslip angle of a vehicle, which also provides the evolution of the uncertainty associated with the estimation. Simulation tests are conducted in Carsim to validate the practical feasibility of the proposed estimator.

II. VEHICLE MODEL

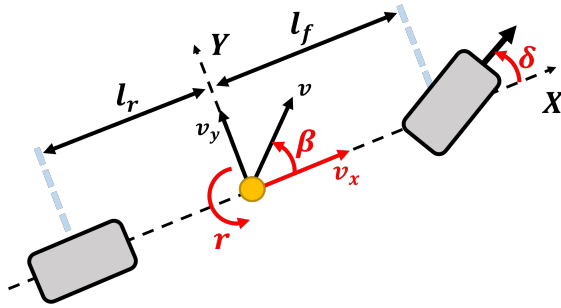


Fig. 1. Vehicle model.

TABLE I
VEHICLE PARAMETERS.

Parameter	Value	Description
l_f	1.14 [m]	Front axle distance to CoG
l_r	1.64 [m]	Rear axle distance to CoG
$C_{\alpha f}$	85–95 [kN/rad]	Front cornering stiffness
$C_{\alpha r}$	60–70 [kN/rad]	Rear cornering stiffness
I_z	4607 [kg·m ²]	Yaw moment of inertia
M	1530 [kg]	Vehicle mass

A nonlinear single-track vehicle model [34] (see **Figure 1**) is selected for designing the sideslip angle estimator. Model parameters can be found in **Table I**. We consider small angles assumption and a slow speed variation. The vehicle dynamics are described by:

$$\dot{x}(t) = \mathcal{A}(v_x)x(t) + \mathcal{B}(v_x)u(t) \quad (1)$$

where β and r are the sideslip angle and yaw rate of the vehicle, respectively. $x = [\beta \ r]^\top \in \mathbb{R}^{n_x}$ is the vehicle state vector. δ is the steering angle. $u = \delta \in \mathbb{R}^{n_u}$ is the control input. The state-space matrices of (1) are

$$\mathcal{A}(v_x) = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathcal{B}(v_x) = \begin{bmatrix} \frac{C_{\alpha f}}{mv_x} \\ \frac{l_s C_{\alpha f}}{I_z} \end{bmatrix}$$

with

$$a_{11} = -\frac{C_{\alpha f} + C_{\alpha r}}{Mv_x}, \quad a_{12} = -\left(1 + \frac{l_f C_{\alpha f} - l_r C_{\alpha r}}{Mv_x^2}\right)$$

$$a_{21} = -\frac{l_f C_{\alpha f} - l_r C_{\alpha r}}{I_z}, \quad a_{22} = -\frac{l_f^2 C_{\alpha f} + l_r^2 C_{\alpha r}}{I_z v_x}$$

It must be noted that system (1) is time-variant, since the longitudinal vehicle velocity v_x changes over time. Furthermore, the cornering stiffness are time-varying due to the complex nature of the tire-road interaction, which depends on several factors. To simplify the design complexity, it is frequent to assume the cornering stiffness bounded; *i.e.*,

$$C_f \in [\underline{C}_f, \overline{C}_f] \\ C_r \in [\underline{C}_r, \overline{C}_r] \quad (2)$$

Applying the Euler's discretization method

$$\dot{x}(t) \simeq \frac{x(k+1) - x(k)}{T}, \quad \forall k \in \mathbb{Z}_+ \quad (3)$$

system (1) is transformed into its discrete-time version

$$x(k+1) = A(v_x)x + B(v_x)u(k) \quad (4)$$

where the discrete-time state-space matrices from (4) are

$$A(v_x) = I + T\mathcal{A}(v_x), \quad B(v_x) = T\mathcal{B}(v_x)$$

and $T = 0.01$ [s] is the sampling time.

It is assumed that the yaw rate r of the vehicle can be measured through an IMU. Then, the measurement vector of the system (4) is

$$y = Cx + F\omega, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad (5)$$

where F and $\omega \in \mathbb{R}^{n_\omega}$ refer to the measurement uncertainty matrix and normalized measurement uncertainty, respectively. $|\omega| \leq 1$. $F = \pi/12$ [rad/s] is selected for this work, in order to ensure that the worst-case inaccuracies are properly addressed.

III. ZONOTOPIC KALMAN FILTER DESIGN

A. Preliminaries

A zonotope $\langle c, R \rangle \subset \mathbb{R}^n$ with the center $c \in \mathbb{R}^n$ and the generator matrix $R \in \mathbb{R}^{n \times p}$ is a polytopic set defined as the linear image of the unit hypercube $[-1, +1]^p \subset \mathbb{R}^p$ by R :

$$\langle c, R \rangle = \{c + Rs, \|s\|_\infty \leq 1\} \quad (6)$$

The Minkowski sum of two sets S_1 and S_2 is $S_1 \oplus S_2 = \{s_1 + s_2, (s_1, s_2) \in S_1 \times S_2\}$. The linear image of the set $S \subset \mathbb{R}^n$ by $L \in \mathbb{R}^{q \times n}$ is $L \odot \langle c, R \rangle = \langle Lc, LR \rangle$. Zonotopes are a class of polytopic sets implicitly represented by matrices,

and they allow efficient set computations. This class is closed under Minkowski sum $\oplus$ and linear image $\odot$:

$$\begin{aligned}\langle c_1, R_1 \rangle \oplus \langle c_2, R_2 \rangle &= \langle c_1 + c_2, [R_1 \ R_2] \rangle \\ L \odot \langle c, R \rangle &= \langle Lc, LR \rangle \\ \langle c, R \rangle \subset \langle c, b(R) \rangle, \quad b(R) &= \text{diag}(|R|\mathbf{1})\end{aligned}\quad (7)$$

where diag returns a diagonal matrix from a vector of diagonal elements and $\mathbf{1}$ is a column vector of ones.

Zonotopic operations result in the generator matrix increasing the column size over time, which will lead to a high computational cost. To address this issue, a reduction operator is typically employed [35]. This involves enclosing the original zonotope within a more conservative one whose generator matrix has fewer columns. By reducing the number of columns, this approach significantly improves computational efficiency while maintaining a bounded approximation of the original set. One possible operator is denoted $\downarrow_{q,W}(\cdot)$, where $q \geq n$ denotes the maximum number of columns for the generator matrix. The reduction operator must satisfy the inclusion property $\langle c, R \rangle \subset \langle c, \downarrow_{q,W}(R) \rangle$. As proposed by Combastel [36], a possible solution is that the reduction operator $\downarrow_{q,W}(\cdot)$ first sorts the columns of $R \in \mathbb{R}^{n \times p}$ according to the decreasing order of the weighted vector norm $\|\cdot\|_W$ and enclosing the generator such that

$$\begin{aligned}R &= [r_1 \ \dots \ r_j \ \dots \ r_p], \quad \|r_j\|_W^2 \geq \|r_{j+1}\|_W^2 \\ p \leq q &\rightarrow \downarrow_{q,W}(R) = R \\ p > q &\rightarrow \downarrow_{q,W}(R) = [R_{>} \ b(R_{<})] \in \mathbb{R}^{n \times q} \\ R_{>} &= [r_1 \ \dots \ r_{q-n}], \quad R_{<} = [r_{q-n+1} \ \dots \ r_p]\end{aligned}\quad (8)$$

B. Observer structure

Following the discrete state-space vehicle model (4) and the measurement vector (5), we have the uncertain time-varying discrete-time system

$$\begin{aligned}x(k+1) &= A(k)x(k) + B(k)u(k) \\ y(k) &= Cx(k) + F\omega(k), \quad \omega_k \in \langle 0, I_{n_\omega} \rangle \subset \mathbb{R}^{n_\omega} \\ x(0) &\in \langle c(0), R(0) \rangle \subset \mathbb{R}^{n_x}\end{aligned}\quad (9)$$

By introducing an observer gain G in (9),

$$\begin{aligned}x(k+1) &= A(k)x(k) + B(k)u(k) \\ &\quad + G(k)(y(k) - Cx(k) - F\omega(k))\end{aligned}\quad (10)$$

The purpose of G is to change the modeling of uncertainty propagation by means of weighting the relative influence of the measurement uncertainty ω_k .

Assuming $x \in \langle c, R \rangle$ at k , it comes:

$$\begin{aligned}x \in \langle c, R \rangle &\implies x(k+1) \in \langle c(k+1), R(k+1) \rangle = \\ &((A - GC) \odot \langle c, \downarrow_{q,W}(R) \rangle) \oplus (B \odot \langle u, 0 \rangle) \\ &\oplus (-GF \odot \langle 0, I_{n_\omega} \rangle)\end{aligned}\quad (11)$$

Now, introduce some covariation matrices:

$$\begin{aligned}P &= RR^\top, \quad Q_\omega = FF^\top, \\ \bar{R} &= \downarrow_{q,W}(R), \quad \bar{P} = \bar{R} \bar{R}^\top\end{aligned}\quad (12)$$

Then, the optimal observer gain G^* is given by

$$\begin{aligned}G^* &= AK^*, \quad K^* = LS^{-1} \\ L &= \bar{P}C^\top, \quad S = C\bar{P}C^\top + Q_\omega\end{aligned}\quad (13)$$

The proof can be found in [37] and is omitted here for simplicity.

Remark 1: Given that the work in [37] assumes the state-space matrices A and B are fully known over time, the present work differs by explicitly addressing model uncertainty in both matrices and analyzing its impact on the evolution of the zonotope during the estimation process.

Remark 2: The calculations in (11 - 13) involve advanced zonotopic operations, such as computing the image of the product of an uncertain matrix and a zonotope. Although this image is generally non-convex, it can be conservatively enclosed by another zonotope. The CORA toolbox [38] provides efficient methods for performing such operations, making it a practical and effective tool for this purpose. As a result, zonotope calculus remains a robust and viable framework for handling estimation uncertainty in this context.

IV. SIMULATION RESULTS

In order to validate the proposed estimator, a series of co-simulations were carried out using MATLAB and CarSim, taking advantage of their integration for high-fidelity vehicle dynamics modeling. The technical specifications of the laptop computer used in this research are as follows: intel i9-12900HX, 32 GB RAM 5600 MHz.

Given that vehicle control lies beyond the scope of this work, the vehicle velocity and steering are controlled by the internal trajectory tracking controller provided by CarSim.

Two illustrative driving scenarios were considered. The first scenario involves a double lane-change maneuver, which is a common practice in everyday driving. Such maneuvers may be performed to avoid obstacles or to overtake slower vehicles. The second scenario utilizes the standard racetrack environment provided within CarSim, offering a more dynamic and challenging setting to assess the estimator's performance under aggressive and nonlinear driving conditions. The road friction coefficient is set as 0.85 in both scenarios. The vehicle trajectory in these scenarios is illustrated in **Figures 2-3**.

The estimator is initialized under the assumption of steady-state vehicle motion; *i.e.*,

$$\begin{aligned}c(0) &= \mathbf{0} \\ R(0) &= \mathbf{0}\end{aligned}\quad (14)$$

where $\mathbf{0}$ denotes a column matrix of zeros.

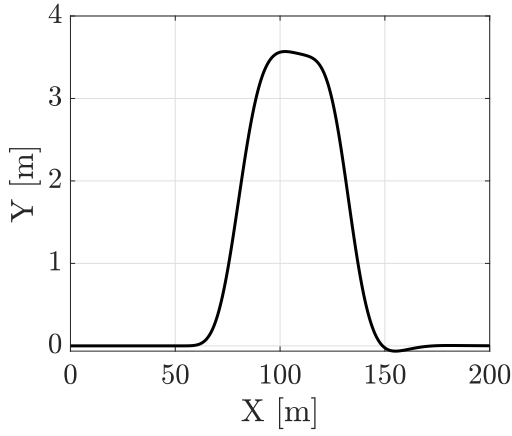


Fig. 2. Vehicle trajectory during the double lane change maneuver.

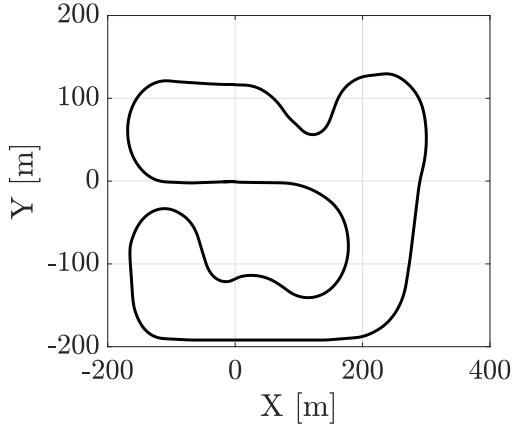


Fig. 3. Vehicle trajectory during the racetrack test.

The vehicle sideslip angle estimation $\hat{\beta}$ obtained with the proposed estimator is depicted in **Figures 4-5**. The validity of the proposed estimator is confirmed by the fact that the zonotope consistently encloses the ground truth value of the sideslip angle throughout the simulation.

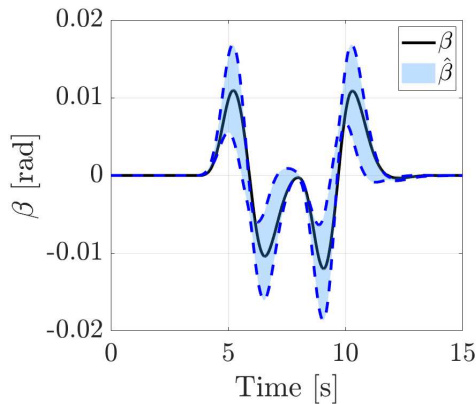


Fig. 4. Sideslip angle estimation during the double lane change maneuver.

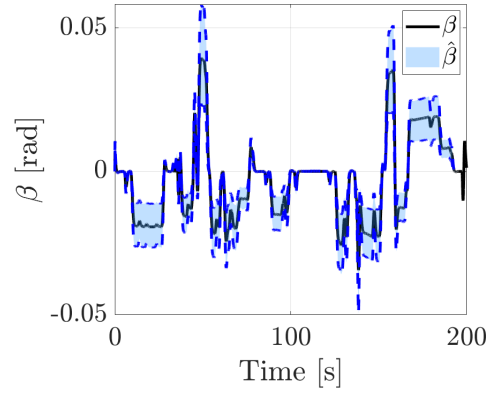


Fig. 5. Sideslip angle estimation during the racetrack test.

V. CONCLUSIONS

This paper has proposed a vehicle sideslip angle estimator based on a zonotopic Kalman filter. The proposed method differentiates itself from existing approaches by explicitly quantifying the uncertainty associated with the sideslip angle estimation. This is achieved by accounting for both measurement noise and model uncertainties. These uncertainties are rigorously propagated through the estimation process, resulting in bounded estimates that provide both a central value and a guaranteed confidence region.

To validate the proposed estimator, a series of co-simulations were performed using CarSim, a high-fidelity vehicle dynamics simulator. Integrated with MATLAB for estimator implementation, this setup enabled testing under realistic conditions. The results demonstrate that, despite accounting for uncertainty, the estimator consistently envelops the ground truth sideslip angle, thereby evidencing its robustness and reliability.

In future works, the proposed system for estimating vehicle sideslip angle and its associated uncertainty will be integrated into a path or trajectory tracking controller, which will also be based on zonotopes. This approach aims to ensure that the vehicle's maximum deviation from the desired path remains within the track limits under bounded uncertainties. Additionally, we aim to design zonotopic observers for fault detection in vehicle platoons, accounting for external disturbances, sensor and actuator faults, network disconnections and false data injection that may compromise platoon stability [39]–[45].

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REFERENCES

- [1] C.-Y. Chan, "Advancements, prospects, and impacts of automated driving systems," *International journal of transportation science and technology*, vol. 6, no. 3, pp. 208–216, 2017.

- [2] M. Gyllenhammar, G. Rodrigues de Campos, and M. Törngren, "The road to safe automated driving systems: A review of methods providing safety evidence," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 4, pp. 4315–4345, 2025.
- [3] T. Huang, R. Fu, Q. Sun, Z. Deng, S. Huang, and L. Jin, "Toward human-vehicle collaboration for automated vehicles: A review and perspective," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 5, pp. 5649–5673, 2025.
- [4] J. Han, J. Zhang, C. Lv, C. He, H. Wei, and S. Zhao, "Robust fault tolerant path tracking control for intelligent vehicle under steering system faults," *IEEE Transactions on Intelligent Vehicles*, pp. 1–11, 2024.
- [5] F. Viadero-Monasterio, M. Meléndez-Useros, N. Zhang, H. Zhang, B. L. Boada, and M. J. L. Boada, "Motion planning and robust output-feedback trajectory tracking control for multiple intelligent and connected vehicles in unsignalized intersections," *IEEE Transactions on Vehicular Technology*, pp. 1–13, 2025.
- [6] Q. Sun, X. Qi, D. Du, X. Ren, and M. Fei, "Event-triggered output feedback model predictive control for path following of autonomous vehicles under false data injection attacks," *IEEE Transactions on Intelligent Vehicles*, pp. 1–13, 2024.
- [7] F. Viadero-Monasterio, M. Jiménez-Salas, M. Meléndez-Useros, H. Zhang, B. L. Boada, and M. J. L. Boada, "Enhancing driving safety through robust static output-feedback path tracking and emergency braking control under network-induced delays," in *2025 International Conference on Control, Automation and Diagnosis (ICCAD)*, pp. 1–6, 2025.
- [8] F. Viadero-Monasterio, A.-T. Nguyen, J. Lauber, M. J. L. Boada, and B. L. Boada, "Event-triggered robust path tracking control considering roll stability under network-induced delays for autonomous vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 12, pp. 14743–14756, 2023.
- [9] S. Zhang, V. Puig, and S. Ifqir, "Robust lpv fault diagnosis using the set-based approach for autonomous ground vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 9078–9090, 2024.
- [10] B. L. Boada, F. Viadero-Monasterio, H. Zhang, and M. J. L. Boada, "Simultaneous estimation of vehicle sideslip and roll angles using an integral-based event-triggered h_∞ observer considering intravehicle communications," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 4, pp. 4411–4425, 2023.
- [11] M. Meléndez-Useros, F. Viadero-Monasterio, M. Jiménez-Salas, and M. J. L. Boada, "Active steering fault diagnosis via integrated lstm-based sensor detection and robust actuator fault estimation," *Reliability Engineering & System Safety*, p. 111573, 2025.
- [12] F. Viadero-Monasterio, J. García, M. Meléndez-Useros, M. Jiménez-Salas, B. L. Boada, and M. J. López Boada, "Simultaneous estimation of vehicle sideslip and roll angles using an event-triggered-based iot architecture," *Machines*, vol. 12, no. 1, p. 53, 2024.
- [13] D. Chindamo, B. Lenzo, and M. Gadola, "On the vehicle sideslip angle estimation: a literature review of methods, models, and innovations," *applied sciences*, vol. 8, no. 3, p. 355, 2018.
- [14] D. Puscul, C. Lex, M. Vignati, and L. Shao, "A literature survey on sideslip angle estimation using vehicle dynamics based methods," *IEEE Access*, vol. 12, pp. 70263–70277, 2024.
- [15] G. Heon Lee, D.-H. Kim, J. Min Pak, and C. K. Ahn, "Vehicle sideslip angle estimation using finite memory estimation and dynamics/kinematics model fusion based on neural networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 2, pp. 2157–2168, 2025.
- [16] L. Gao, Q. Wu, Y. He, K. Wu, and P. Shao, "Robust estimation of sideslip angle for heavy-duty vehicles under payload conditions using a series-connected structure estimator," *IEEE Transactions on Intelligent Vehicles*, pp. 1–17, 2024.
- [17] F. D. Biase, B. Lenzo, and F. Timpone, "Vehicle sideslip angle estimation for a heavy-duty vehicle via extended kalman filter using a rational tyre model," *IEEE Access*, vol. 8, pp. 142120–142130, 2020.
- [18] X. Ding, Z. Wang, and L. Zhang, "Event-triggered vehicle sideslip angle estimation based on low-cost sensors," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 7, pp. 4466–4476, 2022.
- [19] A. Bertipaglia, B. Shyrokau, M. Alirezai, and R. Happee, "A two-stage bayesian optimisation for automatic tuning of an unscented kalman filter for vehicle sideslip angle estimation," in *2022 IEEE Intelligent Vehicles Symposium (IV)*, pp. 670–677, 2022.
- [20] R. Marotta, S. Strano, M. Terzo, and C. Tordella, "On the prediction of the sideslip angle using dynamic neural networks," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 5, pp. 281–295, 2024.
- [21] Z. Huang, "Application of interval state estimation in vehicle control," *Alexandria Engineering Journal*, vol. 61, no. 1, pp. 911–916, 2022.
- [22] F. Viadero-Monasterio, M. Meléndez-Useros, B. Lenzo, and B. L. Boada, "Interval observer for vehicle sideslip angle estimation using extended kalman filters," *Machines*, vol. 13, no. 8, p. 707, 2025.
- [23] Y. Wang, D. M. Bevly, and R. Rajamani, "Interval observer design for lpv systems with parametric uncertainty," *Automatica*, vol. 60, pp. 79–85, 2015.
- [24] C. Combastel, "A state bounding observer based on zonotopes," in *2003 European Control Conference (ECC)*, pp. 2589–2594, 2003.
- [25] V. T. H. Le, C. Stoica, T. Alamo, E. F. Camacho, and D. Dumur, *Zonotopes: From guaranteed state-estimation to control*. John Wiley & Sons, 2013.
- [26] T. Alamo, J. M. Bravo, and E. F. Camacho, "Guaranteed state estimation by zonotopes," *Automatica*, vol. 41, no. 6, pp. 1035–1043, 2005.
- [27] J. J. Glunt, J. A. Robbins, J. A. Siefert, D. Silvestre, and H. C. Pangborn, "Sharp hybrid zonotopes: Set operations and the reformulation-linearization technique," *IEEE Control Systems Letters*, pp. 1–1, 2025.
- [28] V. Puig, P. Cugueró, and J. Quevedo, "Worst-case state estimation and simulation of uncertain discrete-time systems using zonotopes," in *2001 European Control Conference (ECC)*, pp. 1691–1697, 2001.
- [29] H. Zheng, Y. Li, L. Zheng, and E. Hashemi, "Koopman-based hybrid modeling and zonotopic tube robust mpc for motion control of automated vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 10, pp. 13598–13612, 2024.
- [30] D. Ma, T. Ma, Y. Li, and Q. Miao, "Constrained zonotope terrain-aided navigation method for long-range autonomous underwater vehicles," *IEEE/ASME Transactions on Mechatronics*, pp. 1–13, 2025.
- [31] C. Conejo, V. Puig, B. Morcego, F. Navas, and V. Milanés, "A zonotopic fdi approach with lpv-based ekf in autonomous vehicles," *IFAC-PapersOnLine*, vol. 58, no. 4, pp. 443–448, 2024.
- [32] E. Alcalá, V. Puig, J. Quevedo, and O. Senname, "Fast zonotope-tube-based lpv-mpc for autonomous vehicles," *IET Control Theory & Applications*, vol. 14, no. 20, pp. 3676–3685, 2020.
- [33] T. J. Bird, H. C. Pangborn, N. Jain, and J. P. Koeln, "Hybrid zonotopes: A new set representation for reachability analysis of mixed logical dynamical systems," *Automatica*, vol. 154, p. 111107, 2023.
- [34] A.-T. Nguyen, J. Rath, T.-M. Guerra, R. Palhares, and H. Zhang, "Robust set-invariance based fuzzy output tracking control for vehicle autonomous driving under uncertain lateral forces and steering constraints," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 9, pp. 5849–5860, 2021.
- [35] X. Yang and J. K. Scott, "A comparison of zonotope order reduction techniques," *Automatica*, vol. 95, pp. 378–384, 2018.
- [36] C. Combastel, "A state bounding observer for uncertain non-linear continuous-time systems based on zonotopes," in *Proceedings of the 44th IEEE Conference on Decision and Control*, pp. 7228–7234, 2005.
- [37] C. Combastel, "Zonotopes and kalman observers: Gain optimality under distinct uncertainty paradigms and robust convergence," *Automatica*, vol. 55, pp. 265–273, 2015.
- [38] M. Althoff, "An introduction to cora 2015," in *Proc. of the workshop on applied verification for continuous and hybrid systems*, pp. 120–151, 2015.
- [39] F. Viadero-Monasterio, M. Meléndez-Useros, M. Jiménez-Salas, and B. L. Boada, "Robust adaptive control of heterogeneous vehicle platoons in the presence of network disconnections with a novel string stability guarantee," *IEEE Transactions on Intelligent Vehicles*, pp. 1–13, 2025.
- [40] F. Viadero-Monasterio, M. Meléndez-Useros, M. Jiménez-Salas, and B. L. Boada, "Robust adaptive heterogeneous vehicle platoon control based on disturbances estimation and compensation," *IEEE Access*, vol. 12, pp. 96924–96935, 2024.
- [41] F. Viadero-Monasterio, M. Meléndez-Useros, M. Jiménez-Salas, and M. J. L. Boada, "Fault-tolerant robust output-feedback control of a vehicle platoon considering measurement noise and road disturbances," *IET Intelligent Transport Systems*, vol. 19, no. 1, p. e70007, 2025.
- [42] C. Liu, Z. Xia, and R. J. Patton, "Distributed fault-tolerant consensus control of vehicle platoon systems with dos attacks," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 10, pp. 14438–14449, 2024.
- [43] F. Viadero-Monasterio, M. Meléndez-Useros, H. Zhang, B. L. Boada, and M. J. L. Boada, "Signalized traffic management optimizing energy efficiency under driver preferences for vehicles with heterogeneous

powertrains,” *IEEE Transactions on Consumer Electronics*, pp. 1–1, 2025.

- [44] D. A. Godinho, A. Alves Neto, L. A. Mozelli, and F. de Oliveira Souza, “A strategy for traffic safety of vehicular platoons under connection loss and time-delay,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 6, pp. 6627–6638, 2023.
- [45] F. Viadero-Monasterio, M. Meléndez-Useros, M. Jiménez-Salas, B. L. Boada, and M. J. L. Boada, “What are the most influential factors in a vehicle platoon?,” in *2024 IEEE International Conference on Evolving and Adaptive Intelligent Systems (EAIS)*, pp. 1–7, 2024.